AI for SMEs — Handbook

> A living handbook that helps regional manufacturing & wholesale SMEs put AI to work — written by HAN students, one practical question at a time.

> This is the handbook's template. The structure is the shell every cohort fills. Content comes from the teams: researched in the AIBS module, and produced through the research platform the same teams build in AEL.

Start here

Orientation for an SME owner who is new to AI: what this handbook is, who it is for, how to use it, and what it deliberately does not do. Written for a busy non-technical reader — plain language, no jargon, and short enough to read in one sitting.

This handbook is designed to help SMEs like [Company_1] B.V., which saw significant revenue growth of 14.8% in Q3 2026 through AI-driven strategies such as launching an automated logistics platform and expanding into the Scandinavian market. It addresses critical questions about AI adoption, use cases, responsible AI practices, and practical steps for implementation.

The handbook is organized by themes, with each chapter posing a specific question that SME owners might ask themselves. The structure ensures clarity and relevance to regional manufacturing and wholesale contexts. Each theme builds upon previous knowledge gaps, guiding the reader through a structured learning process from basic understanding to advanced application of AI technologies.

The questions this handbook answers

#	Theme	The SME-facing question
1	SME Knowledge Gaps	What do we not know yet about AI, and what would we need to know to decide well?
2	AI Use Cases at SMEs	What do firms like mine use AI for, and what actually pays off?

Chapter template: Theme 1 — What do we not know yet about AI, and what would we need to know to decide well?

One sentence on what this chapter helps an owner do.

Team: AIBS Cohort | SME Context: Dutch Wholesale & Distribution (e.g. sports teamwear) | Sources last appraised: October 2026

1. The question, for this kind of SME

Narrow down the question to regional SMEs (10-100 employees, limited IT staff, seasonal stock spikes).

2. The translation

Turn academic theory into plain guidance an owner can act on. Clearly separate what the firm says it needs from what empirical evidence shows it actually lacks.

3. Illustrative use case (CRISP-DM)

Examine the gap through all six CRISP-DM phases:

- Business understanding: Operational objective (reducing pre-season stockouts).
- Data understanding: Recognizing unorganized historical sales in ERP and broken size runs.
- Data preparation: Time and local cleaning needed before modeling.
- Modelling: Baseline rule-based logic vs. predictive ML.
- Evaluation: Setting strict accuracy thresholds and human-in-the-loop review.
- Deployment: Practical spreadsheet or ERP export integration.

4. Recommended tooling

Concrete, accessible options (no-code first, Excel Power Query, simple Python scripts) with effort and cost estimates.

5. Responsible AI & trust

Data ownership, local data boundaries (no private ERP records in public models), and GDPR compliance.

6. First five steps for an owner

Concrete Monday-morning actions (audit staff AI tools, spot-check ERP data, define a single bottleneck, consult peer SMEs).

7. Sources

Trace claims to field interviews and secondary reports (CBS, Eurostat, trade literature).

Chapter template: Theme 2 — What do firms like mine use AI for, and what pays off?

One sentence on what this chapter helps an owner do.

Team: AIBS Cohort | SME Context: Regional SME Operations & Wholesale | Sources last appraised: October 2026

1. The question, for this kind of SME

Narrow down to practical, high-ROI AI adoption for inventory, customer communications, or forecasting.

2. The translation

How regional SMEs can implement AI pragmatically through 3 maturity tiers:

- Tier 1: Off-the-shelf generative & administrative tools (low effort, quick wins).
- Tier 2: Workflow & task automation (ERP connectors, semi-automated categorization).
- Tier 3: Predictive operations (demand balancing and replenishment alerts).

3. Illustrative use case (CRISP-DM)

Walk through a concrete implementation across the 6 CRISP-DM stages:

- Business understanding: Target ROI and service level improvement.
- Data understanding: Transaction logs and seasonal demand cycles.
- Data preparation: Merging sales history with supplier lead times.
- Modelling: Time-series / decision trees vs baseline moving average.
- Evaluation: Forecast error reduction and stockout reduction metrics.
- Deployment: Automated weekly reorder alert generated for the warehouse manager.

4. Recommended tooling

Accessible tooling index, estimated setup effort in hours, and subscription costs.

5. Responsible AI & trust

Safeguards, bias detection, and human validation for all automated purchase suggestions.

6. First five steps for an owner

Five concrete steps to trial a pilot project within 30 days.

7. Sources

Appraised field research sources, vendor reality check, and industry benchmarks.

Tooling index

The accessible, mostly no-code tools referenced across the chapters, in one place: what each is for, what it costs in money and effort, and which chapter shows it in use.

#	Tool Name	For What	Effort (hrs)	Cost ($)
1	Excel Power Query	Data integration & cleaning	5-10	$0
2	Simple Python Scripts	Basic data analysis and automation	3-7	$0-$50
3	ERP Connectors	Automated workflows within existing systems	8-15	$50-$200

Regional ecosystem & support

Where an owner in the region can go next: innovation centres, hubs, training, funding, and regional networks (e.g. Oost NL, VNO-NCW, HAN Applied Research).

Innovation Centres

HAN Applied Research Centre for AI and Data Science offers workshops, seminars, and one-on-one consultations.

Training & Education

Regional training programs provided by local universities like HAN University of Applied Sciences.

Funding Opportunities

Grants and loans available through regional innovation funds and government grants.

Regional Networks

Networking events and meetups organized by industry associations such as Oost NL and VNO-NCW.

About this handbook

- Who writes it: Students of HAN University of Applied Sciences, in teams, as the integrated task of the AIBS module and the AEL module.
- The quality bar: Clarity, accuracy against appraised sources, responsible-AI coverage, and usefulness.
- Educational, not consultancy: Inspirational guidance, not commercial liability.
- Confidentiality: Real SME cases are anonymised.